

Scenario

As a member of the Security Blue team, your assignment is to analyze a memory dump using Redline and Volatility tools. Your goal is to trace the steps taken by the attacker on the compromised machine and determine how they managed to bypass the Network Intrusion Detection System (NIDS). Your investigation will identify the specific malware family employed in the attack and its characteristics. Additionally, your task is to identify and mitigate any traces or footprints left by the attacker.

What is the name of the suspicious process?

To identify the suspicious process, I used `pslist` and `pstree` plugins to view process list and noticed a `oneetx.exe` which is running from temp directory. Legitimate softwares never run from temp directories rather from standard windows directories like `system32`.

8264	824	RuntimeBroker.	0xad818e8c8080	4	-	1	False	2023-05-21 22:40:33.000000 UTC	N/A	Disabled
3608	676	svchost.exe	0xad818d07a080	3	-	0	False	2023-05-21 22:41:28.000000 UTC	N/A	Disabled
6644	824	SkypeApp.exe	0xad818d3ac080	49	-	1	False	2023-05-21 22:41:52.000000 UTC	N/A	Disabled
5656	824	RuntimeBroker.	0xad81876e8080	0	-	1	False	2023-05-21 21:58:19.000000 UTC	2023-05-21 22:02:01.000000 UTC	
8952	824	TextInputHost.	0xad818e6db080	10	-	1	False	2023-05-21 21:59:11.000000 UTC	N/A	Disabled
5808	824	HxTsr.exe	0xad818de5d080	0	-	1	False	2023-05-21 21:59:58.000000 UTC	2023-05-21 22:07:45.000000 UTC	
2388	5328	msedge.exe	0xad818e54c340	18	-	1	False	2023-05-21 22:05:35.000000 UTC	N/A	Disabled
6292	5328	msedge.exe	0xad818d7a1080	20	-	1	False	2023-05-21 22:06:15.000000 UTC	N/A	Disabled
3876	448	taskhostw.exe	0xad8189b30080	8	-	1	False	2023-05-21 22:08:02.000000 UTC	N/A	Disabled
372	824	SkypeBackgroun	0xad8186f49080	3	-	1	False	2023-05-21 22:10:00.000000 UTC	N/A	Disabled
1120	676	MsMpEng.exe	0xad818945c080	12	-	0	False	2023-05-21 22:10:01.000000 UTC	N/A	Disabled
6076	824	ShellExperienc	0xad818eb18080	14	-	1	False	2023-05-21 22:11:36.000000 UTC	N/A	Disabled
7336	824	RuntimeBroker.	0xad818e8bb080	2	-	1	False	2023-05-21 22:11:39.000000 UTC	N/A	Disabled
7964	5328	msedge.exe	0xad818de5080	19	-	1	False	2023-05-21 22:22:09.000000 UTC	N/A	Disabled
6544	5328	msedge.exe	0xad818c0ea080	18	-	1	False	2023-05-21 22:22:35.000000 UTC	N/A	Disabled
5964	676	svchost.exe	0xad818ef86080	5	-	0	False	2023-05-21 22:27:56.000000 UTC	N/A	Disabled
8896	5328	msedge.exe	0xad8187a39080	18	-	1	False	2023-05-21 22:28:21.000000 UTC	N/A	Disabled
5156	5328	msedge.exe	0xad818c553080	14	-	1	False	2023-05-21 22:28:22.000000 UTC	N/A	Disabled
5896	8844	oneetx.exe	0xad818d143080	5	-	1	True	2023-05-21 22:30:56.000000 UTC	N/A	Disabled
7732	5896	rundll32.exe	0xad818d1912c0	1	-	1	True	2023-05-21 22:31:53.000000 UTC	N/A	Disabled
6324	1496	audiodg.exe	0xad818df2e080	4	-	0	False	2023-05-21 22:42:56.000000 UTC	N/A	Disabled
2228	3580	FTK Imager.exe	0xad818d143080	10	-	1	False	2023-05-21 22:43:56.000000 UTC	N/A	Disabled
5636	3580	notepad.exe	0xad818db45080	1	-	1	False	2023-05-21 22:46:50.000000 UTC	N/A	Disabled
2044	676	svchost.exe	0xad8189b27080	28	-	0	False	2023-05-21 22:49:29.000000 UTC	N/A	Disabled
8708	676	svchost.exe	0xad818d431080	5	-	0	False	2023-05-21 22:57:33.000000 UTC	N/A	Disabled
5476	676	svchost.exe	0xad818e752080	9	-	0	False	2023-05-21 22:58:08.000000 UTC	N/A	Disabled
6596	676	TrustedInstall	0xad818dc88080	4	-	0	False	2023-05-21 22:58:13.000000 UTC	N/A	Disabled
2332	824	TlWorker.exe	0xad818e780080	4	-	0	False	2023-05-21 22:58:13.000000 UTC	N/A	Disabled
4340	676	VSSVC.exe	0xad818e888080	3	-	0	False	2023-05-21 23:01:06.000000 UTC	N/A	Disabled
7540	824	smartscreen.ex	0xad818e893080	14	-	1	False	2023-05-21 23:02:26.000000 UTC	N/A	Disabled
8920	3580	FTK Imager.exe	0xad818ef81080	20	-	1	False	2023-05-21 23:02:28.000000 UTC	N/A	Disabled
5480	448	oneetx.exe	0xad818d3d6080	6	-	1	True	2023-05-21 23:03:00.000000 UTC	N/A	Disabled

**** 4224	6724	Outline.exe	0xad818e8b0800	0	-	1	True	2023-05-21 22:36:23.000000 UTC	2023-05-21 23:01:24.000000 UTC	\Device\HarddiskVolume3\Program Files (x86)\Outline\Outline.exe	-	-
**** 4628	6724	tun2socks.exe	0xad818d823400	0	-	1	True	2023-05-21 22:40:10.000000 UTC	2023-05-21 23:01:24.000000 UTC	\Device\HarddiskVolume3\Program Files (x86)\Outline\resources\app.asar.unpacked\third_party\outline-go-tun2socks\win32\tun2socks.exe	-	-
**** 5536	3580	notepad.exe	0xad818db45080	1	-	1	False	2023-05-21 22:46:50.000000 UTC	N/A	\Device\HarddiskVolume3\Windows\System32\notepad.exe	-	-
**** 464 3580	3580	SecurityHealth	0xad818b908000	3	-	1	False	2023-05-21 22:31:59.000000 UTC	N/A	\Device\HarddiskVolume3\Windows\System32\SecurityHealthSystray.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
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**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
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**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
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**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
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**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
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**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
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**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098000	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	\Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	-	-
**** 5328	3580	msedge.exe	0xad818d098									

Answer: oneetx.exe

What is the child process name of the suspicious process?

Viewing results of `ps tree` revealed that the suspicious process `oneetx.exe` is spawning a child process named `rundll32.exe`. This is common in malwares, they spawn child processes to escalate privileges or for evasion purposes.

4424	6724	Outline.exe	0xadd19e080000	0	-	1	True	2023-05-21 22:16:23.000000 UTC	2023-05-21 23:01:24.000000 UTC	Device\HarddiskVolume3\Program Files (x86)\Outline\Outline.exe	
4428	6724	ts2socks.exe	0xadd18e082340	0	-	1	True	2023-05-21 22:40:10.000000 UTC	2023-05-21 23:01:24.000000 UTC	Device\HarddiskVolume3\Program Files (x86)\Outline\resources\app.asar.unpacked\third_party\outline-go-ts2socks\win32\ts2socks.exe	
5328	3580	netopds.exe	0xadd18d045080	1	-	1	False	2023-05-21 22:16:50.000000 UTC	N/A	Device\HarddiskVolume3\Windows\System32\netopds.exe	
5464	3580	SecurityHealth.exe	0xadd189709080	3	-	1	False	2023-05-21 22:31:59.000000 UTC	N/A	Device\HarddiskVolume3\Windows\System32\SecurityHealth\Sys tray.exe	
5328	3580	msedge.exe	0xadd18d099080	54	-	1	False	2023-05-21 22:32:02.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe"
exe	-	-	-	-	-	-	-	-	-	-	-
5328	3580	msedge.exe	0xadd18d099080	14	-	1	False	2023-05-21 22:32:39.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
8896	5328	msedge.exe	0xadd187a39080	18	-	1	False	2023-05-21 22:28:21.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
5156	5328	msedge.exe	0xadd18c535080	14	-	1	False	2023-05-21 22:28:22.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
7964	5328	msedge.exe	0xadd18c535080	14	-	1	False	2023-05-21 22:32:49.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
4396	5328	msedge.exe	0xadd180515080	7	-	1	False	2023-05-21 22:32:19.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
6544	5328	msedge.exe	0xadd18c0e4080	18	-	1	False	2023-05-21 22:22:35.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
2388	5328	msedge.exe	0xadd18e43c080	18	-	1	False	2023-05-21 22:16:55.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
620	5328	msedge.exe	0xadd180515080	20	-	1	False	2023-05-21 22:32:49.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
1144	5328	msedge.exe	0xadd1875f5080	18	-	1	False	2023-05-21 22:32:38.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
5340	5328	msedge.exe	0xadd18d701080	18	-	1	False	2023-05-21 22:32:39.000000 UTC	N/A	Device\HarddiskVolume3\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	
2152	3580	vmtoolsd.exe	0xadd1893600	8	-	1	False	2023-05-21 22:31:59.000000 UTC	N/A	Device\HarddiskVolume3\Program Files\VMware\VMware Tools\vmtoolsd.exe	"C:\Program Files\VMware\VMware Tools\vmtoolsd.exe" -n vmusr
2228	3580	FTK Imager.exe	0xadd18d143080	18	-	1	False	2023-05-21 22:14:56.000000 UTC	N/A	Device\HarddiskVolume3\Program Files\AccessData\FTK Imager\FTK Imager.exe	
8920	3580	FTK Imager.exe	0xadd18d143080	20	-	1	False	2023-05-21 22:02:28.000000 UTC	N/A	Device\HarddiskVolume3\Program Files\AccessData\FTK Imager\FTK Imager.exe	"C:\Program Files\AccessData\FTK Imager\FTK Imager.exe"
860	588	fontdrvhost.exe	0xadd18761f140	5	-	1	False	2023-05-21 22:27:33.000000 UTC	N/A	Device\HarddiskVolume3\Windows\System32\fontdrvhost.exe	
5996	8844	onextx.exe	0xadd189d41080	5	-	1	True	2023-05-21 22:30:56.000000 UTC	N/A	Device\HarddiskVolume3\Users\Tammam\AppData\Local\Temp\39312af059\onextx.exe	
7322	5986	cmd.exe	0xadd18912c0	1	-	1	True	2023-05-21 22:31:53.000000 UTC	N/A	Device\HarddiskVolume3\Windows\System32\cmd.exe	
parrot	parrot	l - f\vmshare\win									

PPID	Parent Process	Child Process	PID
5896	oneetx.exe	rundll32.exe	7732

Answer: rundll32.exe

What is the memory protection applied to the suspicious process memory region?

To view the memory protection of suspicious process, I first used `vadinfo` to view detailed view of memory region associated with the process. In results, I noticed the entry which has protections `PAGE_EXECUTE_READWRITE`, which is not common. A private region with `RWX` permissions is a strong indicator of injected or self-modifying code.

Progress: 100.00		PDB scanning finished									
PID	Process Offset	Start VPN	End VPN Tag	Protection	CommitCharge	PrivateMemory	Parent	File	File output		
5896	oneetx.exe	0xfffffd818dbdc2c40	0x73da0a00	0x74024fff	Vad	PAGE_EXECUTE_WRITECOPY	554	0	0x0	Windows\SysWow64\Layers.dll	Disabled
5896	oneetx.exe	0xfffffd818dbdc2c40	0x2e60ffff	0x2e60ffff	VadS	PAGE_READONLY	0	0	0xfffffd818dbdc2c40	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc5b70	0x13800000	0x13800000	VadS	PAGE_READWRITE	10	1	0xfffffd818dbdc5200	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc5b70	0x10000000	0x11fffff	VadS	PAGE_READWRITE	11	1	0xfffffd818dbdc5b70	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc1c00	0xc0c0000	0xfbc0fff	Vad	PAGE_READONLY	0	0	0xfffffd8dbdc2b00	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc7ac0	0x8c09000	0xfbc0fff	Vad	PAGE_EXECUTE_WRITECOPY	0	0	0xfffffd818dbdb1c0	\Users\Tammam\AppData\Local\Temp\c3912af058\oneetx.exe	Disabled
5896	oneetx.exe	0xfffffd818dbdc5940	0x4000000	0x437ffff	VadS	PAGE_EXECUTE_READWRITE	56	1	0xfffffd818dc27ac0	N/A	Disabled
5896	oneetx.exe	0xfffffd818ddade20	0x4c00000	0xf0c0fff	Vad	PAGE_READONLY	0	0	0xfffffd818dbdb1c0	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdb9340	0xf000000	0xf40cfff	Vad	PAGE_READONLY	0	0	0xfffffd818ddade20	N/A	Disabled
5896	oneetx.exe	0xfffffd81876f1b30	0xf0f0000	0xf37ffff	Vad	PAGE_READONLY	0	0	0xfffffd818ddade20	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc6b940	0x13600000	0x1361fff	VadS	PAGE_READWRITE	2	1	0xfffffd818dbdc2b00	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc55c0	0x12200000	0x125ffff	VadS	PAGE_READWRITE	11	1	0xfffffd818dbdc6b940	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc2cf00	0x121c0000	0x121cfff	VadS	PAGE_READONLY	0	1	0xfffffd818dbdc55c0	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc6b10	0x12600000	0x135ffff	VadS	PAGE_READWRITE	0	1	0xfffffd818dbdc55c0	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc840	0x13700000	0x1370fff	Vad	PAGE_READONLY	0	0	0xfffffd818dbdc6b940	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdae780	0x13800000	0x1380fff	Vad	PAGE_READONLY	0	0	0xfffffd818dbdae840	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc5c0	0x15100000	0x160ffff	VadS	PAGE_READWRITE	218	1	0xfffffd818dbdc5b70	N/A	Disabled
5896	oneetx.exe	0xfffffd818f029c00	0x14900000	0x149dfff	VadS	PAGE_READWRITE	2	1	0xfffffd818dbdc560	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc8f00	0x12b0ffff	0x12bffff	Vad	PAGE_READWRITE	0	0	0xfffffd818f029c00	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc8e0	0x13800000	0x1380fff	Vad	PAGE_READONLY	0	0	0xfffffd818dbdae800	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc24c0	0x13c00000	0x1488fff	Vad	PAGE_READONLY	0	0	0xfffffd818dbdae800	\Windows\System32\locale.nls	Disabled
5896	oneetx.exe	0xfffffd818e502260	0x14b00000	0x14b7fff	Vad	PAGE_READONLY	0	0	0xfffffd818f029c00	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc3ea0	0x14a00000	0x1424fff	Vad	PAGE_EXECUTE_WRITECOPY	0	0	0xfffffd818e502260	Windows\SysWow64\sfsc.dll	Disabled
5896	oneetx.exe	0xfffffd818dbdc62e0	0x15000000	0x1500fff	Vad	PAGE_READWRITE	0	0	0xfffffd818e502260	N/A	Disabled
5896	oneetx.exe	0xfffffd818f029f00	0x141ffff	0x141ffff	VadS	PAGE_READWRITE	11	1	0xfffffd818dbdc62e0	N/A	Disabled
5896	oneetx.exe	0xfffffd818f029600	0x16c00000	0x16cffff	VadS	PAGE_READWRITE	1	1	0xfffffd818dbdc62e0	N/A	Disabled
5896	oneetx.exe	0xfffffd818f02f600	0x16900000	0x1696fff	VadS	PAGE_READWRITE	1	1	0xfffffd818f029600	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc5d40	0x16b00000	0x16b0fff	Vad	PAGE_READONLY	0	0	0xfffffd818f02f600	N/A	Disabled
5896	oneetx.exe	0xfffffd818e503480	0x16d00000	0x18cffff	Vad	PAGE_READONLY	0	0	0xfffffd818f029600	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc5160	0x18d00000	0x1a50fff	Vad	PAGE_READONLY	0	0	0xfffffd818e503480	N/A	Disabled
5896	oneetx.exe	0xfffffd818f035500	0x23f00000	0x23f0fff	VadS	PAGE_READONLY	1	1	0xfffffd818dbdc5160	N/A	Disabled
5896	oneetx.exe	0xfffffd818f02a190	0x37700000	0x3777fff	VadS	PAGE_READWRITE	1	1	0xfffffd818f035500	N/A	Disabled
5896	oneetx.exe	0xfffffd818dbdc7400	0x35f00000	0x35f6fff	Vad	PAGE_READONLY	0	0	0xfffffd818f02a190	\Windows\Registration\R000	

#	Start VPN	End VPN	Size	Tag	Protection	Commit	Private	File
1	0x73da0000	0x74024fff	~4 MB	Vad	PAGE_EXECUTE_WRITECOPY	554	0	\\Windows\\SysWow64\\AcLayers.dll
2	0x1a60000	0x2e60fff	~20 MB	Vad	PAGE_READONLY	0	0	N/A
3	0x1390000	0x139ffff	64 KB	VadS	PAGE_READWRITE	10	1	N/A
4	0x1000000	0x11ffff	2 MB	VadS	PAGE_READWRITE	11	1	N/A
5	0xfc0000	0xfc0fff	4 KB	Vad	PAGE_READONLY	0	0	N/A
6	0xec0000	0xfb7fff	~1 MB	Vad	PAGE_EXECUTE_WRITECOPY	0	0	\\Users\\Tannan\\AppData\\Local\\Temp\\c3912af058\\oneetx.exe
7	0x400000	0x437fff	224 KB	VadS	PAGE_EXECUTE_READWRITE	56	1	N/A
8	0xfe0000	0xfe0fff	4 KB	Vad	PAGE_READONLY	0	0	N/A

malfind confirmed the region was suspicious.

```

[parrot@parrot ~]$ python3 ../../volatility3/vol.py -f MemoryDump.mem windows.malfind --pid 5896
Volatility 3 Framework 2.28.2
/home/parrot/volatility3/volatility3/framework/deprecation.py:28: FutureWarning: This API (volatility3.plugins.windows.malware.malfind.Malfind.run) will
ugin has been renamed, please call volatility3.plugins.windows.malware.malfind.Malfind rather than volatility3.plugins.windows.malfind.Malfind.
  warnings.warn()

PID    Process Start VPN    End VPN Tag    Protection    CommitCharge    PrivateMemory    File output    Notes    Hexdump Disasm
/home/parrot/volatility3/volatility3/framework/deprecation.py:105: FutureWarning: This plugin (volatility3.plugins.windows.malfind.Malfind) has been rena
6-07. Please ensure all method calls to this plugin are replaced with calls to volatility3.plugins.windows.malware.malfind.Malfind
  warnings.warn()

5896   oneetx.exe      0x400000      0x437fff      VadS    PAGE_EXECUTE_READWRITE  56      1      Disabled      MZ header
4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 MZ.....
b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 .....@.....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 .....
0x400000:    dec    ebp
0x400001:    pop    edx
0x400002:    nop
0x400003:    add    byte ptr [ebx], al
0x400005:    add    byte ptr [eax], al
0x400007:    add    byte ptr [eax + eax], al
0x40000a:    add    byte ptr [eax], al
[parrot@parrot ~]$

```

Answer: PAGE_EXECUTE_READWRITE

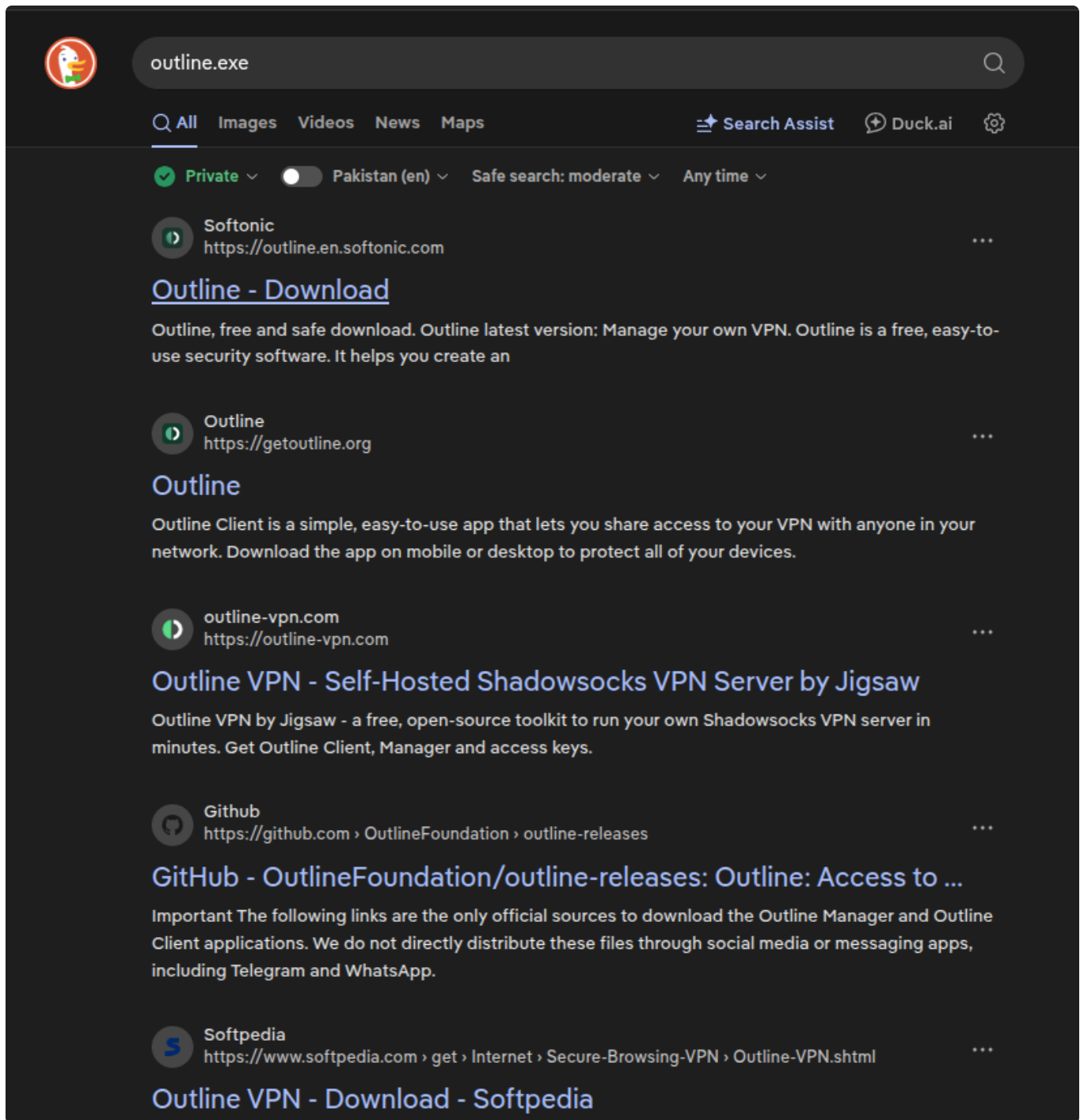
- What is the name of the process responsible for the VPN connection?
- What is the attacker's IP address?

For these tasks, I used netscan plugin to view the network connections. I could not find the process which is responsible for VPN connections. But, I noticed that oneetx had been connected with a IP. Connection is out-bound which means it is C2 endpoint.

```
parrot@parrot: /usr/share/kali$ python3 ../../volatility3/vol.py -f MemoryDump.mem windows.netscan
Volatility 3 Framework 2.28.2
Progress: 100.00
PDB scanning finished
Offset Proto LocalAddr LocalPort ForeignAddr ForeignPort State PID Owner Created
0xad81861e2310 TCPv4 0.0.0.0 49668 0.0.0.0 0 LISTENING 1840 spoolsv.exe 2023-05-21 22:28:09.000000 UTC
0xad81861e2310 TCPv6 :: 49668 :: 0 LISTENING 1840 spoolsv.exe 2023-05-21 22:28:09.000000 UTC
0xad81861e2470 TCPv4 0.0.0.0 5040 0.0.0.0 0 LISTENING 1196 svchost.exe 2023-05-21 22:30:31.000000 UTC
0xad81861e2730 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 952 svchost.exe 2023-05-21 22:27:36.000000 UTC
0xad81861e2b50 TCPv4 0.0.0.0 49665 0.0.0.0 0 LISTENING 552 wininit.exe 2023-05-21 22:27:36.000000 UTC
0xad81861e2b50 TCPv6 :: 49665 :: 0 LISTENING 552 wininit.exe 2023-05-21 22:27:36.000000 UTC
0xad81861e2e10 TCPv4 0.0.0.0 49665 0.0.0.0 0 LISTENING 552 wininit.exe 2023-05-21 22:27:36.000000 UTC
0xad81861e3230 TCPv4 0.0.0.0 49664 0.0.0.0 0 LISTENING 696 lsass.exe 2023-05-21 22:27:36.000000 UTC
0xad81861e3390 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 952 svchost.exe 2023-05-21 22:27:36.000000 UTC
0xad81861e3390 TCPv6 :: 135 :: 0 LISTENING 952 svchost.exe 2023-05-21 22:27:36.000000 UTC
0xad81861e34f0 TCPv4 0.0.0.0 49664 0.0.0.0 0 LISTENING 696 lsass.exe 2023-05-21 22:27:36.000000 UTC
0xad81861e34f0 TCPv6 :: 49664 :: 0 LISTENING 696 lsass.exe 2023-05-21 22:27:36.000000 UTC
0xad81861e37b0 TCPv4 0.0.0.0 49666 0.0.0.0 0 LISTENING 1012 svchost.exe 2023-05-21 22:27:49.000000 UTC
0xad81861e37b0 TCPv6 :: 49666 :: 0 LISTENING 1012 svchost.exe 2023-05-21 22:27:49.000000 UTC
0xad81861e3910 TCPv4 0.0.0.0 49667 0.0.0.0 0 LISTENING 448 svchost.exe 2023-05-21 22:27:58.000000 UTC
0xad81861e3910 TCPv6 :: 49667 :: 0 LISTENING 448 svchost.exe 2023-05-21 22:27:58.000000 UTC
0xad81861e3a70 TCPv4 0.0.0.0 49668 0.0.0.0 0 LISTENING 1840 spoolsv.exe 2023-05-21 22:28:09.000000 UTC
0xad81861e3bd0 TCPv4 0.0.0.0 49666 0.0.0.0 0 LISTENING 1012 svchost.exe 2023-05-21 22:27:49.000000 UTC
0xad81861e3e90 TCPv4 0.0.0.0 49667 0.0.0.0 0 LISTENING 448 svchost.exe 2023-05-21 22:27:58.000000 UTC
0xad818662ecb0 TCPv4 0.0.0.0 445 0.0.0.0 0 LISTENING 4 System 2023-05-21 22:29:04.000000 UTC
0xad818662ecb0 TCPv6 :: 445 :: 0 LISTENING 4 System 2023-05-21 22:29:04.000000 UTC
0xad818662f390 TCPv4 0.0.0.0 7680 0.0.0.0 0 LISTENING 5476 svchost.exe 2023-05-21 22:58:09.000000 UTC
0xad818662f390 TCPv6 :: 7680 :: 0 LISTENING 5476 svchost.exe 2023-05-21 22:58:09.000000 UTC
0xad81878518f0 UDPv4 192.168.190.141 138 * 0 4 System 2023-05-21 22:27:56.000000 UTC
0xad8187852250 UDPv4 192.168.190.141 137 * 0 4 System 2023-05-21 22:27:56.000000 UTC
0xad818902a5d0 TCPv4 192.168.190.141 139 0.0.0.0 0 LISTENING 4 System 2023-05-21 22:27:56.000000 UTC
0xad818971f870 UDPv4 0.0.0.0 56250 * 0 6644 SkypeApp.exe 2023-05-21 22:58:07.000000 UTC
0xad818971f870 UDPv6 :: 56250 * 0 6644 SkypeApp.exe 2023-05-21 22:58:07.000000 UTC
0xad81897eb010 TCPv4 10.0.85.2 55439 20.22.207.36 443 CLOSED 448 svchost.exe 2023-05-21 23:00:40.000000 UTC
0xad81898a6d10 UDPv4 127.0.0.1 57787 * 0 448 svchost.exe 2023-05-21 22:28:54.000000 UTC
0xad81898bc7f0 UDPv4 0.0.0.0 5355 * 0 1448 svchost.exe 2023-05-21 22:57:37.000000 UTC
0xad81898bc7f0 UDPv6 :: 5355 * 0 1448 svchost.exe 2023-05-21 22:57:37.000000 UTC
0xad8189a291b0 TCPv4 0.0.0.0 55972 0.0.0.0 0 LISTENING 5964 svchost.exe 2023-05-21 22:27:57.000000 UTC
```

Proto	Local Addr	Local Port	Foreign Addr	Foreign Port	State	PID	Owner
TCPv4	10.0.85.2	55462	77.91.124.20	80	CLOSED	5896	oneetx.exe

For the process with VPN connection, I went through the process list again and noticed `outline.exe`. After searching, it was confirmed that this process was VPN utility indeed.



Answer (Q4): `outline.exe`

Answer (Q5): `77.91.124.20`

What is the full URL of the PHP file that the attacker visited?

For this, I used `strings` command with regular expression to list all the urls ending with `php` and one of them was the one attacker visited.

```
strings MemoryDump.mem | grep -E "https?:/*.*\.php"
```

```

[parrot@parrot]--[~/vmshare/kde]
$strings MemoryDump.mem | grep -E "https?:/?.*\.(php|exe)"
nfi!.http://209.11.244.51/p.php?n=
=http://gbfhju.com/r.php></
https://thl9.tk/office-dream.php',
https://msn.com https://msn.com https://hbx.media.net/cksync.php?type=nms&cs=3&ovsid=2159D74C474D6B9A2450C45B46E76A80&gdpr=0&gdpr_consent=
, "http://9ygw2.com/iz5/yaca.php
://spr-updates.ddns.net/spr_updates.php"-q-nhttp://spr-updates.ddns.net/spr_updates.php-o
https://www.ludashi.com/cms/server/monitor.php?id=
http://dongtaiwang.com/loc/phome.php?v=
http://dongtaiwang.com/loc/phome.php?v=
=http://gbfhju.com/r.php></
http://skdjui.com/r.php>
https://thl9.tk/office-dream.php',
http://77.91.124.20/store/games/index.php
2src="http://%s/js.php?affid
lehttp://9bgng.com/iz5/yaca.php
http://skdjui.com/r.php>
http://dongtaiwang.com/loc/phome.php?v=
http://dongtaiwang.com/loc/phome.php?v=
2src="http://%s/js.php?affid
http://77.91.124.20/store/games/index.php
http://77.91.124.20/store/games/index.php
=http://gbfhju.com/r.php></
http://skdjui.com/r.php>
http://yuoioip.info/rd/rd.phphttp://nanoatom.info/rd/rd.php{8d5849a2-93f3-429d-ff34-260a2068897c}
[parrot@parrot]--[~/vmshare/kde]
$

```

Answer: `http://77.91.124.20/store/games/index.php`

What is the full path of the malicious executable?

In the initial analysis, we could notice the path from which `oneetx.exe` was executed when viewing `pstree` results.

Answer: `C:\Users\Tammam\AppData\Local\Temp\c3912af058\oneetx.exe`

Summary

A compromised Windows host was analyzed using Volatility 3 revealing a dropper-style malware infection. The suspicious process `oneetx.exe` was found running from a user temp directory (`C:\Users\Tammam\AppData\Local\Temp\c3912af058\oneetx.exe`) where it spawned a child process `rundll32.exe` and allocated a private `PAGE_EXECUTE_READWRITE (RWX)` memory region which is a clear indicator of fileless code injection. Network analysis showed the malware connected outbound to `77.91.124.20` over HTTP (port 80) to a `.php` endpoint which was the attacker's C2 server. The attacker bypassed the NIDS using `outline.exe`, a VPN utility that hid the C2 communication from signature-based detection. Key IOCs include the malicious executable path, the attacker IP `77.91.124.20`, the PHP URL and VPN utility.